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A novel ensemble approach based on MCC and MCDM methods for prioritizing tweets mentioning urban issues in smart city

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Abstract

Purpose – Social media platform, like Twitter, has increasingly become the mode of reporting civic issues owing to their vast and fast reachability. A tremendous amount of information on urban issues is shared every moment out of which some tweets may need immediate attention to save lives or avoid future disasters. Existing approaches are only limited to the identification of complaint tweets; however, its prioritization based on urgency is still unexplored. This study aims to decide the ranking of complaints based on its criticality derived using multiple parameters, like type of complaint, season, day or night, gender, holiday or working day, etc.

Design/methodology/approach – The approach proposes an ensemble of multi-class classification (MCC) and "two-level" multi-criteria decision-making (MCDM) algorithms, like AHP and TOPSIS, to evaluate the accurate ranking score of the tweet based on the severity of the issue. Initially, the MCC is applied to tweets to categorize the tweets into three categories, i.e. moderate, urgent and immediate. Further, the first level of MCDM algorithm decides the ranking within each complaint type, and the second level evaluates the ranking across all types. Integration of MCC and MCDM methods further helps to increase the accuracy of the result.

Findings – The paper discusses various parameters and investigates how their combination plays a significant role in deciding the priority of complaints. It successfully demonstrates that MCDM techniques are helpful in generating the ranking score of tweets based on various criteria.

Originality/value – This paper fulfills an identified need to prioritize the complaint tweet which helps the local government to take time-bound actions and save a life.

Keywords Twitter, Multi-class classification, Multi-criteria decision making, Urban issues

Paper type Research paper

1. Introduction

Citizens have become more interactive in sharing the daily life issues they face and use the social media platform, like Twitter, (López-Ornelas *et al.*, 2017) to spread the concern, get the attention to certain inconvenience (Estévez-Ortiz *et al.*, 2016) or even in case of emergency. Due to the easy and quick access of the Twitter platform, it has become a popular medium to seek for the help or to discuss life threating issues, which could be used to avoid fatal accidents in future. Along with the everyday issues, people also mention crucial problems which need to be identified and segregated from the huge amount of secondary issues in order to solve on priority. Ranking of the issues helps to attend and allocate the limited recourse to the correct location and time to avoid any kind of disaster. In one such example (https://fightthenewdrug.org/train-passengers-tweet-saved-girls-from-trafficking/), tweets acted upon in time also helped police to rescue 25 minor girls from the probable human trafficking case.

A continuous stream of tweets makes the task of manually distinguishing the critical issues from the noncritical ones very strenuous and imprecise. So here in this study, we implement the automatic classification of the complaints into three classes, like "moderate," "urgent" and "immediate" using the multi-class classification (MCC) algorithm (Mills, 2018). Instead of the traditional "urgent" and "non-urgent" binary classification, the proposed approach splits the issues into three classes for more precise categorization as manual efforts would be required to identify issues from "urgent" tweets, which need immediate action. The



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class information determines the level of urgency while dealing with the issue. The tweets coming under the "immediate" class need to attend promptly without any further delay whereas "urgent" tweets need to solve as soon as possible though may not be instantly. "Moderate" tweets convey the concern which has the lowest priority as compared to "immediate" and "urgent" tweets and can be handled based on availability of resources.

The process of classification of the complaint tweets into three different types of categories, i.e. moderate, urgent and immediate, helps to improve the response time taken to attend the issues as the issues will be handled according to the class label attached with it. However, considering the volume of complaint tweets and variety of issues, many issues can be labeled as "urgent" or "immediate," which makes it even more difficult to choose the first issue to resolve among many issues specially urgent or immediate. Even though binary classification of the urban issues has been explored before, "ranking" of the tweets based on the severity of the issue is rarely studied, which can play a vital role in decision-making while solving the problems, especially in case of emergency. The severity level of the issue depends upon many parameters, like time of the day, season, gender, holiday or not and many more, for example, open manhole complaint posted in rainy season should have higher priority than the same complaint posted in summer season. Similarly, if another open manhole complaint is posted in the rainy season but at night, it should be prioritized with the higher ranking score than the previous one. In this type of complaint, "gender" parameter does not play a role, so it should not be considered important one. However, in case of "safety" or "crime" related complaints, "gender" and "time of day" parameters hold crucial value rather than "season," etc. So the significance of the parameters or features varies as per the type of complaint. Hence, to evaluate the ranking sequence of the tweets within each category, our study investigates the multi-criteria decision-making (MCDM) techniques, which generate the ranking score based on various criteria. Our approach proposes an ensemble of MCC algorithm and "two-level" MCDM algorithm to evaluate the accurate ranking score of the tweet. The first level of the MCDM algorithm decides the ranking within each complaint type, and the second level evaluates the ranking across all the types. The integration of the MCC result to the MCDM result further helps to increase the accuracy of the result. Our approach tried to prioritize the complaint tweets based on qualitative attributes, like gender, time, season, type of complaint etc. rather than existing quantitative measures, which will help local government to take time-bound actions and save life.

To the best of our knowledge, this is the first study to implement an ensemble of MCC and MCDM techniques to rank individual tweet based on the severity of the issue mentioned in it.

2. Related works

Recent research has shown that researchers have explored various methods to analyze the tweets and arrange them according to their urgency. Alshehri (2019) has presented an integrated framework based on three approaches, like clustering, classification and ranking, to identify the tweets sent during disaster and rank them as urgent or not. Here the author relied on the supervised machine learning algorithm to classify the priority of the tweets. A dictionary of around 400 most frequent disaster-related words was created and utilized as a feature in deciding the urgency of the tweet. Out of many classified urgent tweets, which need to be attended first still remains as a manual task. Masdeval and Veloso (2015) calculated the urgency of civic issues by mining different emotions from the text describing the issue. Author has performed the experiments using the SeeClickFix hackathon data and various regression algorithms, like ordinary least square (OLS) and support vector regression (SVR), in order to derive no. of votes for each issue, which based solely on the emotions extracted from its textual description. More the number of the votes, greater will be the urgency. The authors have only considered the emotional aspect of issues ignoring other parameters,

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like type and time of issue, its frequency, recency and many more. Kejriwal and Zhou (2020) utilized local word and pretrained wiki embedding models along with the transfer learning to detect the urgency of the issues mentioned in the short texts. Here local and wiki word embeddings of the labeled and unlabeled data were used as features to the ensemble of classifiers. If weighed score of all regression models is above a predetermined threshold, then the message is flagged as urgent. Gan *et al.* (2020) proposed a forecasting model to predict the severity of the accident utilizing the United Kingdom (UK) road safety dataset. Authors explored the Deep Forests algorithm based on decision tree ensemble. However, this study did not emphases on optimizing the model parameters, which could result in improvement of prediction accuracy. Deshmukh and Shiravale (2018) developed a framework for Pune Municipal Corporation, which will accept the citizen's complaints and provide immediate response to their queries. The authors have experimented with the combination of WordNet knowledge base and machine learning algorithms to generate the responses as well as estimate the grievance level of the issues performing sentiments analysis. Here the authors have only considered sentiment strength as the measure of the urgency and hence given the higher priority to those complaints having greater negative sentiment score. Complaints with less or no negative sentiments but conveying more severe issues can be neglected by this approach. Amora *et al.* (2018) evaluated various machine learning techniques to perform the sentiment analysis of tweets in Brazilian Portuguese to prioritize customer service issues. Results of the experiments indicated the efficiency of deep learning approaches (long short-term memory (LSTM) and bidirectional long short-term memory (BLSTM)) in classification. Instead of relying only on the negative sentiments, authors have categorized the complaints into positive, negative and critical based on the dissatisfaction level. Devaraj *et al.* (2020) developed the framework to classify the tweets requesting help during the Hurricane in 2017. The author has used the convolutional neural network (CNN) model trained on word embeddings along with the traditional natural language processing (NLP) features, like n-gram and part-of-speech (POS) tags adopted for multilayer perceptron. This study experimented with the average word embeddings as features for support vector method (SVM) classifiers to classify urgent tweets and performed well when compared with other proposed approaches. Here while deciding the urgency of the tweets, the scope of the task was limited to specific tweets related to Hurricane Harvey, which should be generalized. Besides the study omitted the location information of the tweets while deciding the emergency level, which could be a prominent factor in addressing the issue. Lee *et al.* (2019) examined unstructured civil complaints for safety and disaster management to understand the policy demands of 16 towns. Authors have used advanced big data analysis techniques, such as text mining, and data fusion and integration, to understand citizen's needs. Instead of social media data, complaints registered on telephone and demographic dataset were used for experimentation purpose. Authors have tried four types of complaints, which are hygiene, lighting facilities, traffic safety and disaster management and utilized Korean Natural Language Processing for the preprocessing of data. Frequency-based analysis of words generated 200 most frequent words for analysis. The relationship between various characteristics of the citizens like age, gender and income rate and the safety complaints was explored to understand the policy demands. Ghodousi *et al.* (2019) studied the citizen's complaint system of the first district of Bojnurd municipality in Iran. The authors have utilized K-means and Bees algorithms separately as well as simultaneously to identify citizen's urgent needs and measure their level of satisfaction. Results of the BA clustering were better due to Silhouette and Calinski-Harabasz indexes. Purohit *et al.* (2018) proposed the learning to rank-based supervised learning method for automatic classification and ranking of the serviceable request. The classification of task decides the request is serviceable or not and ranks the request in the list based on their serviceability index. Burel and Alani (2018) developed an open source web API based on CNN to segregate the social media posts during

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crisis. This API can identify the type of events and provide categories of information as well. The level of urgency of the issues was determined by Bansal and Toshniwal (2016) based on density and co-occurrence of the different type of complaints over a particular time line. The criticality score of the complaint was determined using the linear regression equation based on density and temporal aspect of the complaint.

While studying the prioritization of the tweets, many studies have utilized the dataset related to the natural disasters where many tweets were available mentioning the emergency situation and hence convenient to identify the high priority tweets. Very few studies have discussed on the priority of the tweets discussing the daily issues and can turn into the fatal accidents. Even in most of such studies, sentiment analysis result was considered for classifying the text as "urgent" or "not urgent." The severity of the issue was also derived from the quantitative value which means in order to be the highest priority issue, quantity of tweets mentioning that issue should be very high. In this scenario, one or two tweets commenting on life threatening concern or asking for immediate help will be totally ignored and considered "not urgent" owing to lack of count. Furthermore after classification of tweets, the resultant "urgent" class may consist of many tweets which make it again a manual task to decide the priority of each urgent tweet to identify which issue to attend first. So instead of only binary classification of urban complaints into "urgent" and "not urgent", MCC could provide more precise classification of complaint tweets. Moreover the urgency of the issue cannot be only derived from the sentiment expressed in it but depends upon many factors which themselves may have diverse relevance pertaining to different type of complaints. This kind of scenario can be handled by the popular MCDM method.

2.1 Multi criteria decision-making (MCDM)

It is the prominent technique which evaluates possible alternatives against different conflicting criteria to reach to the final decision. The primary steps to follow in MCDM are as shown in Figure 1.

Even though numerous methods of MCDM are studied by the researchers, some of the popular methods of MCDM along with their specific area of application are discussed by Patel *et al.* (2017). Ozkaya and Erdin (2020) have evaluated various dimensions, like transportation, smart living, smart governing, etc. for number of cities across the world to rank the criteria helping the city to become smarter and manageable. Criteria were weighted and ranked using analytic network process (ANP) and TOPSIS (Technique for Order Preference by Similarity to Ideal Solution). Various indicators having the prominent impact on mobility system in the smart city, like congestion, pollution, safety and accident reduction, were evaluated and ranked (Zapolskytė *et al.*, 2020) using the AHP method.

Transportation has been active area where researchers have studied the use of MCDM methods to grade the transport infrastructure (Broniewicz and Ogrodnik, 2020; Erdogan and

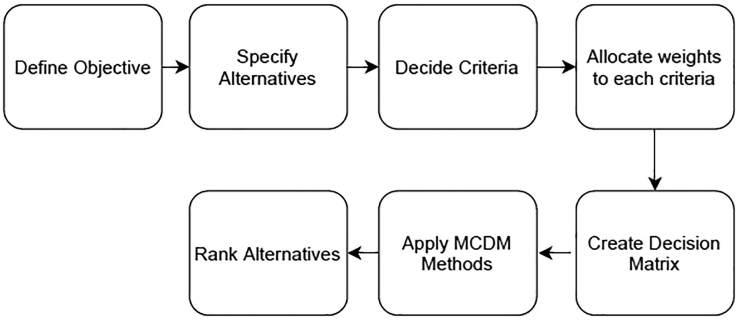


Figure 1.
Basic block diagram of multi criteria decision-making process

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Kaya, 2019; Farooq *et al.*, 2019; Stanković *et al.*, 2019). Farooq *et al.* (2018) explored with AHP MCDM method along with the geographic information system (GIS) data to decide the best transportation alternative. Researchers (Farooq *et al.*, 2021) have also applied GIS and SWOT analysis in addition to based evaluation method (BEM) and PROMETHEE II method to determine the most suitable mode of transport in Peshawar city.

In most of the earlier studies, the AHP has been preferred for solving complex decision-making issues using various criteria. Smart city concept mainly delves into citizens, local government and stakeholders. In order to deliver right services to the citizens with optimized use of the available resources, the local government's decision-makers try to prioritize the needs and solutions to implement smart city product service system (Rondini *et al.*, 2018). The AHP and TOPSIS are popular techniques used for decision-making. The AHP method calculates the weights of each criterion and those weights are used by TOPSIS method to rank the alternatives (Bhutia and Phipon, 2012). In the AHP, problems are decomposed into a hierarchy of criteria and alternatives and weights are derived for each alternative. The TOPSIS method calculates the distance of each alternative from the ideal solution and negative ideal solution (NIS). The aim of this method is to select the alternative, which is closest to ideal solution and farthest from NIS. Table 1 summarizes the investigations for prioritizing tweets mentioning urban issues in smart city with general machine learning techniques and MCDM.

2.2 AHP (analytic hierarchy process)

The AHP (Saaty, 1980) is based on decomposing a complex MCDM problem into a system of hierarchies. There is a fundamental 1–9 scale of absolute numbers shown in Table 2 to design the hierarchy.

AHP method, first proposed by Saaty (1980), is a popular method for evaluating the solution for complex problems and deriving the weights of the decision criteria. It mainly consists of three parts: first is the problem under study, second is the possible solution to the problems called as alternatives and third in various criteria using which alternatives are evaluated to achieve final objective. In this method, the criteria are compared pairwise and quantified to decide the best alternative for the given problem. The AHP method is much easier to understand and implement. The process of calculating the weights of each criteria is described in Figure 2. Suppose the different criteria considered for problem are $C_1, C_2 \ldots, C_n$, and the possible alternatives are $A_1, A_2 \ldots .. A_m$. Every criterion compared with every other criteria for relative importance for achieving the final objective. For comparison purpose, the five-point (1-3-5-7-9) or nine-point (1-2-3-4-5-6-7-8-9) scale as mentioned in Table 2 suggested by Saaty (2008) is used. However, the same scale can be reorganized into three point (1-2-3) or five point scale (1-2-3-4-5) as well.

Using the scale, the experts compare each criterion with other and design the pairwise comparison matrix (CM). The elements of the CM are as shown in Table 3. For the problem in hand, if the criteria C_1 is extremely important and criteria C_3 is slightly important, then C_1/C_3 value will be derived as 9/2 and so on.

Once the CM is formulated, it is normalized by dividing each matrix elements by sum of each criteria value. As the weights or relevance of each criterion is decided by the expert based on his understanding of the problem, evaluation of the compatibility or consistency of the assessments is done by calculating the "consistency ratio (CR)" The CR which determines the degree of compatibility of matrix can be derived using below mentioned formula.

$$C.R. = \frac{C.I.}{R.I.} \tag{1}$$

where CI = Consistency index and RI = Random index.

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Year	Author	Title	Purpose/objective	Summary	Limitation/ observation
2019	Alshehri	A Machine Learning Approach to Predicting Community Engagement on Social Media During Disasters	Identify the tweets sent during disaster and rank them as urgent or not	Framework based on three approaches clustering, classification and ranking	Relied on supervised machine learning algorithm, which needs readily available labeled data. Out of many classified urgent tweets which need to be attended, first still remains as a manual task
2015	Masdeval C. and Veloso A.	Mining citizen emotions to estimate the urgency of urban issues. Information systems	Calculate the urgency of civic issues	Used various regression algorithms like ordinary least square (OLS) and support vector regression (SVR) in order to derive no. of votes for each issues, which is based solely on the emotions extracted from its textual description	The authors have only considered emotional aspect of the issues ignoring other parameters like type and time of issue, its frequency, recency and many more
2020	Kejriwal, M., and Zhou, P.	On detecting urgency in short crisis messages using minimal supervision and transfer learning. Social Network Analysis and Mining	Detect the urgency of the issues mentioned in the short texts	Utilized local word and pretrained wiki embedding models along with the transfer learning. If weighed score of all regression models is above a predetermined threshold, then the message is flagged as urgent	Based on only content of the text ignoring other imp parameters like gender, time etc. in deciding urgency of the issue
2020	Gan, J., Li, L., Zhang, D., Yi, Z. and Xiang, Q.	An alternative method for traffic accident severity prediction: using deep forests algorithm. Journal of advanced transportation	Predict the severity of the accident	Authors explored the Deep Forests algorithm based on decision tree ensemble	This study did not emphases on optimizing the model parameters which could result in improvement of prediction accuracy
2018	Deshmukh, K.V. and Shiravale, S.S.	Priority based sentiment analysis for quick response to citizen complaints	Accept the citizen's complaints and provide immediate response to their queries	The authors have experimented with the combination of WordNet knowledge base and machine learning algorithms to generate the responses as well as estimate the grievance level of the issues performing sentiments analysis	Authors have only considered sentiment strength as the measure of the urgency. Complaints with less or no negative sentiments but conveying more severe issues can be neglected by this approach

Table 1.
Summarized literature survey

(*continued*)

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Year	Author	Title	Purpose/objective	Summary	Limitation/ observation
2018	Amora, P.R.P., Teixeira, E.M., Lima, M.I.V., Amaral, G.M., Cardozo, J.R.A. and de Castro Machado, J.	An Analysis of Machine Learning Techniques to Prioritize Customer Service Through Social Networks	Prioritize customer service issues mentioned in Brazilian Portuguese tweets	Evaluated various machine learning techniques, like LSTM and BLSTM, to perform the sentiment analysis	Complaints were categorized based on only dissatisfaction level
2020	Devaraj, A., Murthy, D. and Dontula, A.	Machine-learning methods for identifying social media-based requests for urgent help during hurricanes	Developed the framework to classify the tweets requesting help during the Hurricane	This study experimented with the average word embeddings along with the traditional NLP features, like n-gram and part-of-speech tags as features for SVM classifiers	Scope of the task was limited to specific tweets related to Hurricane Harvey, which should be generalized. Besides the study omitted the location information of the tweets while deciding the emergency
2019	Lee, E., Lee, S., Kim, K.S., Pham, V.H. and Sul, J.	Analysis of Public Complaints to Identify Priority Policy Areas: Evidence from a Satellite City around Seoul	Examined unstructured civil complaints for safety and disaster management to understand the policy demands	Used advanced big data analysis techniques, such as text mining and data fusion and integration to understand the citizen's needs	Complaints registered on telephone and demographic dataset having generally a fixed format were used for experimentation instead of more complex social media text. Also only four types of complaints were studied
2019	Ghodousi, M., Alesheikh, A. A., Saeidian, B., Pradhan, B., and Lee, C. W.	Evaluating Citizen Satisfaction and Prioritizing Their Needs Based on Citizens	Studied the citizen's complaint system of the first district of Bojnurd municipality in Iran	Utilized K-means and Bees algorithms to identify citizen's urgent needs and measure their level of satisfaction	Other important parameters like gender, time, etc. in deciding urgency of the issue were studied.
2018	Purohit, H., Castillo, C., Imran, M. and Pandey, R.	Social-eoc: Serviceability model to rank social media requests for emergency operation centers	Automatic classification and ranking of the serviceable request	Proposed learning-to-rank-based supervised learning method	–
2018	Burel, G. and Alani, H.	Crisis event extraction service (crees)-automatic detection and classification of crisis-related content on social media	Segregate the social media posts during crisis	CNN-based approach	–
2016	Bansal, P. and Toshniwal, D.	Analyzing civic complaints for proactive maintenance in smart city	Determined the urgency of issues	Criticality score of the complaint was determined using linear regression equation based on density and temporal aspect of the complaint	Urgency of the issues determined by density factor which will ignore life-threatening issue mentioned in only few complaint

(*continued*)

Table 1.

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Year	Author	Title	Purpose/objective	Summary	Limitation/ observation
2020	Ozkaya, G. and Erdin, C.	Evaluation of smart and sustainable cities through a hybrid MCDM approach based on ANP and TOPSIS technique	Ranked the criteria helping the city to become smarter and manageable	ANP and TOPSIS	Evaluated the criteria based on various dimensions, like transportation, smart living and smart governing
2020	Zapolskytė, S., Burinskienė, M. and Trepanier, M.	Evaluation criteria of smart city mobility system using MCDM method	Evaluate indicators having the prominent impact on mobility system	Analytic hierarchy process (AHP)	–
2020	Broniewicz, E., and Ogrodnik, K.	Multi-Criteria Analysis of Transport Infrastructure Projects	Grade the transport infrastructure and city services using MCDM techniques	Analytic hierarchy process (AHP)	The AHP has been preferred for solving complex decision-making issues using various criteria
2019	Erdogan, M., and Kaya, I.	Prioritizing Failures by Using Hybrid Multi Criteria Decision Making Methodology With a Real Case Application			
2019	Farooq, A., Xie, M., Stoilova, S., and Ahmad, F.	Multicriteria Evaluation of Transport Plan for High-Speed Rail: An Application to Beijing-Xiongan			
2019	Stanković, M., Gladović, P., and Popović, V.	Determining the Importance of the Criteria of Traffic Accessibility Using Fuzzy AHP and Rough AHP Method			
2018	Rondini, A., Lagorio, A., Pinto, R. and Pezzotta, G.	A multi-criteria decision-making approach for prioritising product-service systems implementation in smart cities			

Table 1.

Importance	Description
1	Equally important
2	Slight importance
3	Moderate importance
4	Moderate to strong importance
5	Strong importance
6	Strong to very strong importance
7	Very strong importance
8	Very strong importance to extreme importance
9	Extreme importance

Table 2.
Nine point scale proposed by Saaty (2008)

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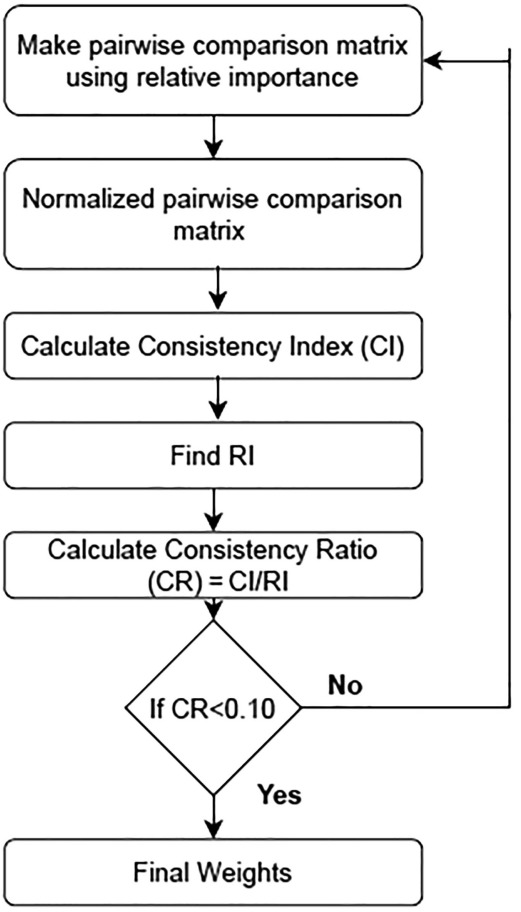


Figure 2.
Evaluating criteria weights using AHP

	$x = 1, 2 \ldots .. n$				
	C_1	C_2	C_3	...	C_x
C_1	1	C_1/C_2	C_1/C_3	...	C_1/C_x
C_2	C_2/C_1	1	C_2/C_3	...	C_2/C_x
C_3	C_3/C_1	C_3/C_2	1	...	C_3/C_x
C_x	C_x/C_1	C_x/C_2	C_x/C_3	...	1

Table 3.
Pairwise CM for criteria

RI has fixed value as mentioned below based on number of columns (n) in matrix.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
R.I.	0	0	0.52	0.89	1.12	1.26	1.36	1.41	1.46	1.49	1.52	1.54	1.56	1.58

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CI (Saaty, 1980) can be determined as follows:

$$C.I. = \frac{(\lambda max - m)}{(m-1)} \tag{2}$$

where λmax = Average (derive ratio of weighted sum and criteria weights of each criteria)
m = no. of alternatives

Lesser the value of CI, better is the CM. Ideal value of CI is "0". RI has predefined values depending upon number of criteria. Once the values of CI and RI are calculated, CR value for the CM can be derived, which is considered compatible when CR score is ≤ 0.1.

2.3 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

The TOPSIS method was first developed in 1981 by Yoon and Hwang (1981). It selects the alternative having shortest distance from the ideal solution and the farthest from the NIS (Yoon, 1980).The major advantages of the TOPSIS method are its simplicity and easy to compute steps with high computational efficiency. The basic steps of TOPSIS method are as mentioned in Figure 3.

Initially, the decision matrix is formed with the values of each alternative for each possible criterion. Afterward, the matrix values are multiplied by resp. criteria weights, and the resultant matrix is further normalized. In the next step positive ideal solution (PIS) and NIS are derived. For each alternative, the distance from the PIS and NIS is determined and followed by its relative closeness to the ideal solution. The alternatives are ranked based on the value of the relative closeness coefficient. The computational procedure of TOPSIS is explained in detail as below.

Suppose there are "m" alternatives for a problem which are $A_1, A_2 \ldots A_m$ and "n" criteria like $C_1, C_2 \ldots C_n$, value V_{ij} shows the value of alternative A_i for criteria C_j where $i = 1 \ldots$ m and $j = 1 \ldots n$. Weights of associated with each criteria is represented as W_k where $\sum_{k=1}^{n} \mathrm{W}_k = 1$

(1) The decision matrix will be like Table 4.

As V_{ij} represents different types of values based on different criteria domain, the decision matrix needs to be normalized. Equations 2, 3 calculate the normalized values, which are further converted into the weighted normalized form using equation 4.

$$N_{ij} = \frac{V_{ij}}{\sqrt{\sum_{i=1}^{m} X_{ij}^2}} \tag{3}$$

$$WN_{ij} = W_j * N_{ij} = W_j * \frac{V_{ij}}{\sqrt{\sum_{i=1}^{m} X_{ij}^2}} \tag{4}$$

(2) The weighted normalized matrix can be plotted as follows:

$$\mathrm{WN} = \begin{Bmatrix} WN_{11} & WN_{12} \ldots\ldots & WN_{1n} \\ WN_{21} & WN_{22} \ldots\ldots & WN_{2n} \\ . & . & . \\ . & . & . \\ . & . & . \\ WN_{m1} & WN_{m2} \ldots\ldots & WN_{mn} \end{Bmatrix} \tag{5}$$

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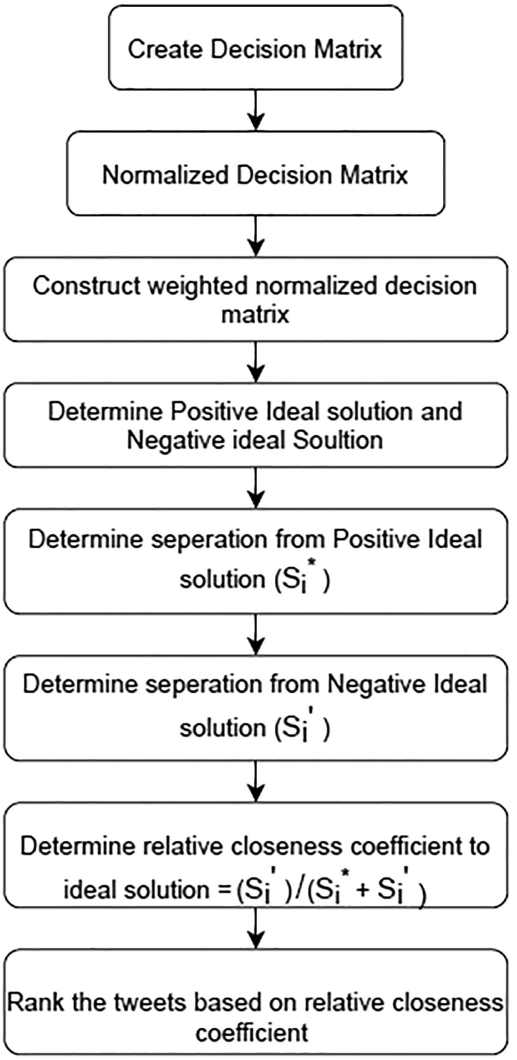


Figure 3.
Basic steps of TOPSIS method

Criteria (weights) → Alternatives↓	C_1 (W_1)	C_2 (W_2)	....	C_n (W_n)
A_1	V_{11}	V_{12}	...	V_{1n}
A_2	V_{21}	V_{22}	...	V_{2n}
...	...	...	...	...
...	...	...	...	...
A_m	V_{m1}	V_{m2}	...	V_{mn}

Table 4.
Decision matrix for TOPSIS method

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(3) The positive and negative ideal solutions are determined as follows:

PIS:

$$A^{+} = \{(\max_{i} V_{ij} \mid j \in J), (\min_{i} V_{ij} \mid j \in J')\} = \{V_1^{+}, V_1^{+}, V_2^{+} \ldots .. V_j^{+}\} \quad (6)$$

Determine NIS:

$$A^{-} = \{(\min_{i} V_{ij} \mid j \in J), (\max_{i} V_{ij} \mid j \in J')\} = \{V_1^{-}, V_1^{-}, V_2^{-} \ldots .. V_j^{-}\} \quad (7)$$

J represents the benefit (maximization).

J' indicates the loss (minimization).

(4) Calculation of separation measure:

$$\text{Positive Ideal Solution} = S_i^{+} = \sqrt{\sum_{j=1}^{n} \left(V_{ij} - V_j^{+}\right)^2} \; i = 1, 2 \ldots .m \quad (8)$$

$$\text{Negative ideal Solution} = S_i^{-} = \sqrt{\sum_{j=1}^{n} \left(V_{ij} - V_j^{-}\right)^2} \; i = 1, 2 \ldots .m \quad (9)$$

(5) Calculating relative closeness to the ideal solution:

$$C_i^{*} = \frac{S_i^{-}}{\left(S_i^{+} + S_i^{-}\right)} \; i = 1, 2 \ldots .m \quad (10)$$

3. Proposed solution approach

In this section, we have described a two-stage model for predicting the priority of the tweets based on its content. Figure 4 represent the proposed framework based on an ensemble of MCC and MCDM for ranking the tweets based on its severity. Initially, the tweets are preprocessed to remove unwanted noise and make the tweet more relevant for classification. In the next step, features are extracted using popular techniques, like TF-IDF or BOW. Some of the important tweets-related features are also extracted from its content, which will play the vital role in achieving the final output. The next step includes application of the MCC techniques, which is a supervised approach where tweets are assigned a relevant label from possible set of labels, i.e. "immediate," "urgent" and "moderate." Simultaneously, the MCDM technique with two different set of criteria and alternatives are explored where weights of the criteria are estimated using the AHP method, and ranking is decided using the TOPSIS method. All the three outcomes, i.e. severity level and ranking scores obtained by two parallel MCDM scenarios, are used to decide the final priority score of the tweets. Detailed description of each of these steps is as follows.

3.1 Preprocessing

The tweets extracted using Twitter streaming API for any study are very raw and unstructured, which consist of many redundancies and unwanted information, so preprocessing of the tweets is an important step of the overall framework which covert the

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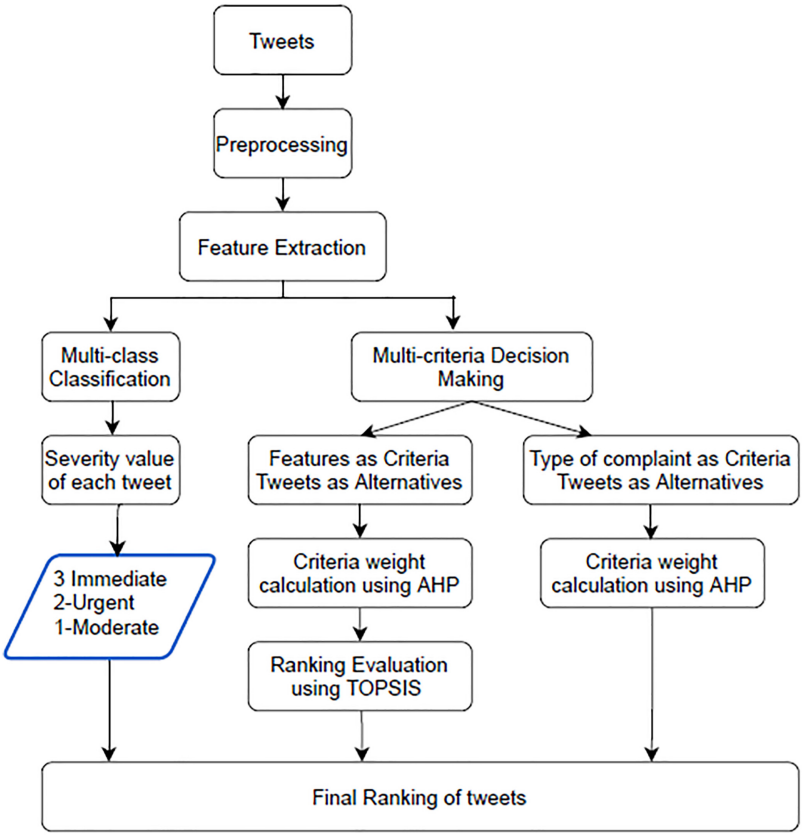


Figure 4.
Two stages complaint priority prediction framework

tweets into the machine readable form for further processing. There are several preprocessing techniques which are specially applicable for the informal nature of the tweets, which are described as follows.

(1) Discarding tweets composed of less than "*n*" number of words – normally very short tweets fail to convey any kind of information, so they become useless for further analysis. Hence, the tweets containing less than or equal to three words are discarded from the training data.

(2) Stop words removal – stop words are the words like articles, conjunction or pronoun like "a," "an," "and," "so," etc. which are commonly present in the sentence but rarely carry any importance in text classification. Thus, these words are filtered so that the overall size of the training data get reduced and hence the training time by keeping only the important words.

(3) Tokenization – instead of experimenting with complete sentence at once, each sentence is divided into subparts and every separate entity is considered a meaningful input for text mining. This process is called as "tokenization," which separates the sentence into tokens. Tokens can be either words or characters as well. It is an important step in NLP where tokens are generated from entire dataset. These

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tokens are further used for creating vocabulary from the corpus which ultimately utilized by the feature extraction steps to extracting the features.

(4) Stemming and lemmatization – widely used in NLP, the process of reducing inflection to the root of the words is called as "stemming." Example of the stemming will be like "played" or "playing" converted to one word "play." However, the resulting root or stem may not be the correct word. Lemmatization also reduces the inflected word to the root word but makes sure that the root word will be correct word present in the language. In order to get the correct root word, i.e. "lemma", POS of the words needs to be defined first.

(5) Part-of-speech (POS) tagging – The POS tagging is the process of identifying the type of word depending on its position in the sentence. It determines whether the word is noun, verb or adjective and so on.

(6) No change of letter cases – in most of existing studies, all the words are converted to the lower case for ease of test mining task. However, as our study is based upon the severity of the situation where while conveying the seriousness of the situation, one can express the problem by using the uppercase letters, which normally shows higher impact than lower case letters. For example, "PLEASE HELP" conveys the strong emotion as compared to "please help," so the words are not converted to lower case and kept as they are.

(7) Retaining particular punctuation symbols – Punctuation symbols like "?" or "!" are used to express the strong emotions e.g. "Please do something, it's request!!!!!," so unlike existing studies, the punctuation symbols are not removed and used for further analysis. Other punctuation symbols, like comma, full stop and so on, are removed from the tweets.

3.2 Feature extraction

As the tweets are written in natural language, in order to apply the machine learning algorithm, the text needs to convert from raw data to features. Machine learning algorithms are trained on these features extracted from training corpus to generate the output for test data. Various types of features used in our study are as follows.

(1) *Bag of words (BOW)*: One of the popular methods of converting text into features is BOW where each word will be considered feature. Here a vector of tokens is created which will be fed to the classification algorithms. Ordering of the words can be preserved by using "n-gram" method mostly bigrams and trigrams instead of unigrams.

(2) *Twitter-specific features*: some of the features are specifically related to the Twitter platform, which can be used for better learning of classification algorithms.

- *No. of followers:* number of followers indicates the popularity of the user. It shows number of people to whom the user's messages are visible. So more the number of followers, more will be the spread of message.
- *User's friend count:* again the user's count of number of friends shows his network on the twitter platform.
- *Count of mentions:* mentions specified by "@username" allow other users to take part in the discussion. More number of mentions in the tweet indicates more involvement of users.

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- *Count of hash tags:* hash tags mainly represent the recent trends or topics being discussed on the twitter platform. It also helps to distinguish the tweets according to different topics.

(3) *Psychometric features*:

- *Fear:* whether the tweets indicated "Fear" emotion(1) or not(0) from its content helps decide the urgency of the tweet as normally fear emotion invokes in some safety or crime related issues.
- *Anger or frustration:* It denotes whether people express their anger/frustration (1) or not (0) can suggest the delayed issues or problems which are causing so much of inconvenience.
- *Request:* Many times users send request using the tweets to solve the any issues they are facing. Some of the request may be generalized while some of users may send the urgent distress request in any of the life threatening situation, which should not be ignored. Hence, this feature denotes whether tweet contains request (1) or not (0).

(4) *Generalized features*:

- *Location present or not:* in order to handle the issue, location information is very important, so this feature indicates the presence (1) or absence (0) of location details in tweet.
- *Count of consecutive question marks and exclamation marks*: in order to communicate strongly, users can use multiple question marks or exclamation marks. Thus, this feature indicates count of consecutive question marks and exclamation marks.
- *Count of capital words:* As the capital words are used to convey the expression more intensity so count of capital words present in the tweet can convey urgency or severity.
- *Day_type:* some of the problems, like traffic, worsen and need to be handled proactively during the weekdays rather than weekends, so the feature "Day_type" will indicate 1 for weekdays and 0 for weekends.
- *Holiday:* normally on the holidays indicated by "0" the number of issues and their intensity are less so it also matters on deciding the priority of the issue. Nonholiday days are indicated by "1".
- *Day_night:* whether the problem is reported on day(0) or night(1) also helps to understand the urgency of the issue as generally the complaints like "safety" if reported on night should handle on priority as compared to the same issues reported on daytime.
- *Season:* urgency of some of the issues depends upon the season whether it is summer, winter or rainy season. Potholes, traffic jams and open manholes like issues could be disastrous specially in rainy season as compared to summer or winter, so the feature value will be "1" if the complaint is posted in rainy season and "0" if the complaint is posted in summer or winter seasons.
- *Gender:* gender of the complainant may not matter in all the issues, like noise pollution and all, but in some complaints, like safety and crime, it plays important role in order to take timely decisions. Here feature value will be "1" for female and "0" for male.

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- *Count of important words:* various words like "help," "danger," "urgent" and so on are probable words which help to distinguished among the more desperate issue, so count of presence of these types of words can be decisive (see Table 5).
- *Type of complaints:* In our study, we have included 14 different complaint types which also contribute towards deciding the priority of the issue. Each of these 14 different types of complaints is considered a feature where the value "1" indicates that the tweet belongs to that complaint category.

3.3 Multi-class classification

In our study, as the tweets are classified into three classes, i.e. "immediate," "urgent" and "moderate," instead of binary classification, traditional MCC algorithm is applied. Various base classification algorithms, like logistic regression and support vector machine, can be used for MCC as well.

3.4 Multi-criteria decision-making

In order to rank the tweets related to urban issues, 14 types of complaints and all the features described in previous section are used as indicators or criteria. Tweets are considered as alternatives for which values of various criteria are evaluated to derive the ranking score. In the AHP method, pairwise CM of all criteria is prepared where each criteria is compared with other criteria in terms of importance for all the alternatives. In our study, the importance of the criteria varies based on each type of complaint, so overall 14 pairwise comparison matrices using all the criteria are formulated for each type of complaint. So in first stage, weights of all the criteria for all 14 types of complaints are evaluated using the AHP method. Furthermore, ranking of the tweets is also based on which type of issue it contains, e.g. safety-related complaint should always be more important than garbage-related issues, so in the second stage, one more pairwise CM is prepared with 14 types of complaints, which are considered criteria. Experts like faculties and students working in social media analytics and NLP area of research are consulted to formulate and finalize the pairwise matrices as importance estimation of the criteria as the rest of the evaluation is based on this step.

The ranking score obtained after employing the weights of the criteria calculated in first stage and applying TOPSIS method gives the score of the tweet within each type of complaint, i.e. if there are 15 tweets discussing the "Manhole" complaint, then ranking of all these 15 complaints based on various criteria will be done in the first stage.

Later on, the final overall ranking of the tweets is evaluated based on its severity score (S), ranking score of within category of complaint ($Rank_{IntraType}$) and weight assigned to the type of complaint to which the tweet belongs ($Wt_{complaint\ category}$).

The final ranking of the tweet is decided by following equation.

$$R = S * Rank_{IntraType} * Wt_{complaint\ category} \tag{11}$$

Table 5. Example of important words

Important words
Urgent, distress, terrible, abuse, please, act, tease, request, misbehave, danger, blackmail, injury, deadliest, dead, fatal, accident, immediately, helpless, risky, die, asap, deep, broken, unsafe, unauthorized, robbery, scary, major, threat, life, help, need, urgency, bad, deeper, worse, deathtrap, mishap, huge, quick, collapse, untoward, stamped, rush, graveyard, harmful, problem, suffer, unsafe, harass, tragedy, destroy

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4. Experimental set up

Twitter has become the popular method to share these problems and convey the discomfort, warning and sometime even request for help during the emergency situation. We have collected the data using the Twitter streaming API using the specific keywords related to the urban issues and important twitter handle like "@MumbaiPolice", "@mybmc," etc. The dataset contains around 20,000 tweets which are filtered to remove irrelevant tweets, and the final set consisted of around 2000 samples. The tweets were labeled into three classes as "3", "2" and "1" which stands for "immediate", "urgent" and "moderate," respectively. Some of the examples of the tweets which can be categorized as "immediate", "urgent" and "moderate" are as mentioned below.

(1) *Immediate:* these tweets mostly were giving warning about some life threatening situation, which need to be attended immediately to safe life. In some cases, even if the situation is not dangerous but causing lots of inconvenience which need to solved at that moment only can also be the part of this class. Examples of these kinds of tweets are as follows.

- *Sample tweet 1*: "@mybmcWardMW Broken Gutter lid on busy Road needs urgent replacement. Location: Next to Hotel Reloaded, Waman Tukaram Patil Marg, Chembur East, Mumbai. Plz replace @mybmc "
- *Sample* tweet 2: "ANOTHER URGENCY. A big ditch right outside Manmala Devi Mandir, Mahim is causing cars lining up till Ganga Vihar circle. This road is already in a bad traffic zone. Pls attend immediately @mybmcRoads @mybmcWardGN @mybmc @MumbaiPolice @RoadsOfMumbai @mumbaitraffic @mumbaimatterz"
- *Sample tweet 3*: "PNR 8723410209 COACH STINKING BADLY UNABLE TO BREATHE. @WesternRly @drmadiwr @RailMinIndia"

(2) *Urgent*: the tweets mentioning the issues which are serious and which may not require immediate action but action as soon as possible were considered as urgent tweets. Some of the examples of urgent tweets are as mentioned follows.

- *Sample tweet 1*: "@MumbaiPolice everyday people stand outside shweta beer shop (near fatima high school vidyavihar west) and they have their drinks there and some of them teasing women plz look into it. Its an request"
- *Sample tweet 2*: "The bridge near Belapur station Navi Mumbai Maharashtra is in a dangerous condition. Plzz Share This Central Railway look in to this plz @mybmc @RailwaySeva @WesternRly @CPMumbaiPolice @RailMinIndia"
- *Sample tweet 3*: "@mybmc @MumbaiMirror @MumbaiPolice @AUThackeray @mybmcWardFN this problem of unfinished footpath has not been fixed even though it was reported in March. Bad footpath has caused many injuries to @pedestrians during this monsoon"

(3) *Moderate*: Some of the issues which were not life threatening, serious and could be resolved in near future were considered moderate issues and examples of tweets consisting of those kinds of issues are as follows:

- *Sample tweet 1*: "@mybmcWardS @mybmc @mybmcRoads Happy to see one more patch work, question is how long will it sustain? Patch work 2 days back gave away in yesterday's rain. I will keep you posted on the status of these potholes. Is the water leakage arrested? @PotholeWar"

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- *Sample tweet 2*: "Lord @mtptraffic @MumbaiPolice you REALLY REALLY REALLY have to do something about the traffic light at Elphinstone. For the AMOUNT of TAX and FINES we pay this is not okay!! IT TAKES MINIMUM 20 min to get through. PLEASE DO SOMETHING!! "
- *Sample tweet 3*: "The entire stretch of J.P. road from Andheri to Versova is filled with illegal political banners for taking mileage during Ganpati immersions. Why no action against them @mybmc! Just because it is election time and you will act only after the 11th day immersion @Dev_Fadnavis"

5. Results and discussion

In this section, we present the results obtained at every step of proposed ensemble approach. Initially, the MCC results are analyzed followed by all the intermediate results achieved at every stage of the MCDM methods. Here results are compared to examine the efficiency of the proposed features as well as two-stage MCDM approach.

5.1 Multi-class classification results

In order to classify the tweets in one of the three classes, like "immediate," "urgent" and "moderate," MCC algorithm is experimented with various base classifiers. All the features as described in section 3.2 were extracted and provided as an input to the multi-class classifiers to get the output in terms of severity class 3, 2 and 1, which stand for "immediate," "urgent" and "moderate" class, respectively. The detailed results of the experiments are as shown in Table 6.

The parameters considered for result comparison are accuracy, macro-precision, macro-recall and macro-F1 score. The macro-average averages the performance of the individual classes instead of overall samples. Greater the value of macro-F1, better is the performance of the classifier for each class, which makes it more suitable for the dataset having imbalanced class distribution. In our research, class imbalance exists as the issues which need immediate attention will be far less as compared to "urgent" and "moderate" issues. In real-life scenario, moderate class issues are reported exceedingly as compared to urgent and immediate issues, so all the results are compared using macro-average parameters.

As depicted in Table 6, most of the multi-class classifier models show better results with the logistic regression algorithm. So, Figure 5–10 depict the ROC curve for Models 1–6, respectively, with LR algorithm which clearly indicates that the classification status improved satisfactorily and performed best with 77% accuracy and 66% macro-F1 score.

5.2 Multi-criteria decision-making results

In this section, ranking of the tweets are done using two scenarios. In both the scenarios, AHP and TOPSIS methods are evaluated using mathematical equations with the procedure explained in literature survey section.

5.2.1 Features based multi-criteria decision-making. In first scenario, features "fear," "ang_fru," "request," "day_type," "holiday," "loc_present," "day_night," "season," and "gender" are considered criteria, and pairwise CM of all the criteria is formulated for "*each complaint type*". As there are 14 types of complaints, 14 matrices are generated in total. The scale [1-2-3-4-5] equivalent to Saaty's nine-point scale is used for the experimentation, as shown in Table 7. For example, for "safety" type of complaint, "gender" was assigned the score "5" as compared to "3" given to the criterion "season" and for "manhole" type of complaint, "season" criterion assigned the score "3" whereas "gender" criterion was assigned with "1" as gender does not matter in case of "manhole" type complaints.

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Sr. no	Features	Algorithm	Acc	Macro-precision	Macro-recall	Macro-F1
Model 1	Preprocessed tweets	MultinomialNB	0.66	0.5	0.53	0.51
		Decision tree	0.64	0.51	0.5	0.5
		Random forest	0.73	0.72	0.51	0.53
		GNB	0.66	0.5	0.53	0.51
		Logistic regression	0.72	0.63	0.54	0.55
Model 2	[Preprocessed Tweets + Psychometric features + Twitter specific features] + Pipeline + grid search	MultinomialNB	0.71	0.59	0.57	0.57
		Decision tree	0.71	0.65	0.53	0.56
		Random forest	0.73	0.57	0.48	0.48
		GNB	0.73	0.47	0.45	0.45
		Logistic regression	0.76	0.71	0.59	0.62
Model 3	[Preprocessed Tweets + Psychometric features + Twitter specific features+ count_imp_words] + Pipeline + grid search	MultinomialNB	0.70	0.56	0.56	0.56
		Decision tree	0.71	0.65	0.56	0.58
		Random forest	0.72	0.65	0.49	0.51
		GNB	0.66	0.47	0.45	0.45
		Logistic regression	0.75	0.69	0.53	0.54
Model 4	[Preprocessed_tweet + Psychometric features + Twitter specific features+ count_imp_words + type_of_complaints] + Pipeline + grid search	MultinomialNB	0.72	0.52	0.54	0.52
		Decision tree	0.77	0.71	0.60	0.62
		Random forest	0.75	0.70	0.52	0.55
		GNB	0.66	0.47	0.45	0.45
		Logistic Regression	0.77	0.68	0.60	0.62
Model 5	[Preprocessed_tweet + Psychometric features + Twitter specific features + Generalized features] + Pipeline + grid search	MultinomialNB	0.74	0.58	0.56	0.54
		Decision tree	0.75	0.69	0.58	0.60
		Random forest	0.77	0.79	0.57	0.61
		GNB	0.66	0.47	0.45	0.45
		Logistic regression	0.76	0.69	0.63	0.65
Model 6	[Preprocessed_tweet + Psychometric features + Generalized features] + Pipeline + grid search	MultinomialNB	0.74	0.60	0.57	0.55
		Decision tree	0.72	0.62	0.56	0.58
		Random forest	0.73	0.65	0.56	0.58
		GNB	0.66	0.47	0.45	0.45
		Logistic regression	*0.77*	0.71	*0.64*	*0.66*

Table 6. Multi-class classification results with various feature combinations

Table 8 shows the example for pairwise CM for issues mentioning the complaints about "Road".

In pairwise CM, every criterion is compared with rest of the criteria using the scale mentioned in Table 7. For "Road" type of complaint, loc_present vs holiday is 5/2 which indicates "loc_present" criterion is of strong importance as compared to slight important "holiday" criterion.

Accordingly, 14 pairwise CMs are generated, and applying the equations from AHP method, weights of the criteria are derived for each type of complaint. Consolidated weights derived for all the criteria for individual "type of complaint" are displayed in Table 9 along with the graphical comparison as shown in Figure 11.

The CR which determines the degree of compatibility of matrix is derived for each of the 14 formulated matrixes based on "type of complaint" using equations 1 and 2 and displayed in Table 10. CR value for the CM is considered compatible when CR score is ≤ 0.1, and the CR values obtained for all the 14 types of complaints are ≤ 0.1 indicating all the weights derived

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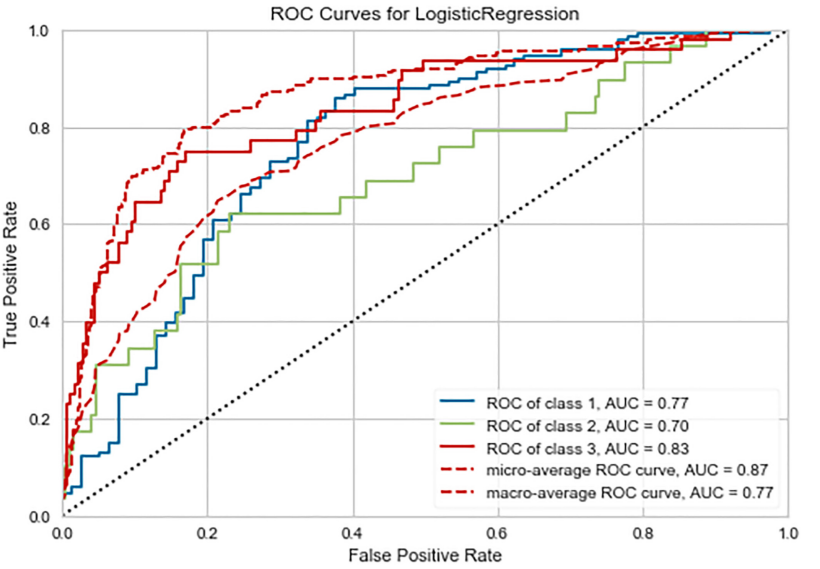


Figure 5.
ROC curve of logistic regression algorithm for Model 1

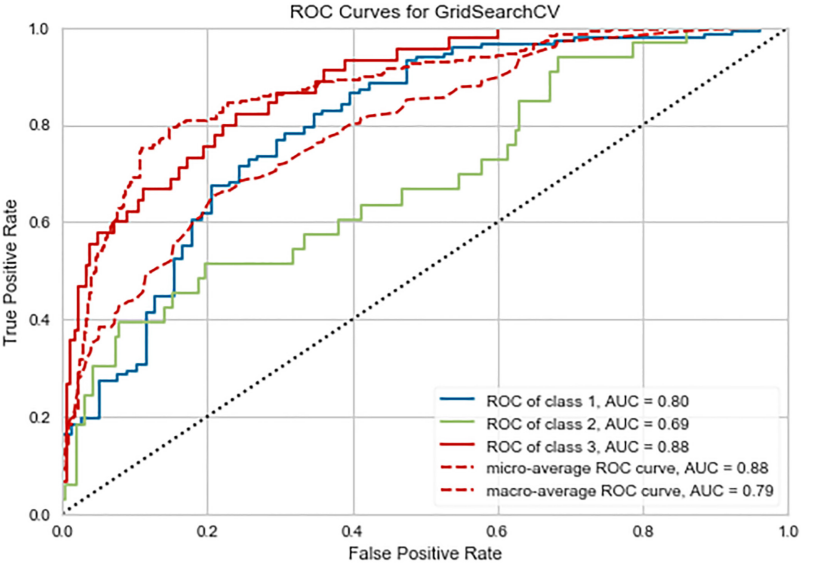


Figure 6.
ROC curve of logistic regression algorithm for Model 2

from the pairwise CM using the equations mentioned in AHP method are appropriate and can be utilized further for calculating overall ranking of tweets using TOPSIS.

Once the weights of the criteria are derived, the TOPSIS method is applied on the tweets considered alternatives, and ranking is estimated. Again here, the TOPSIS method is applied for each "type of complaint," so 14 numbers of iterations are performed to determine the ranking of the tweets belonging to each type. First, the tweets corresponding to each

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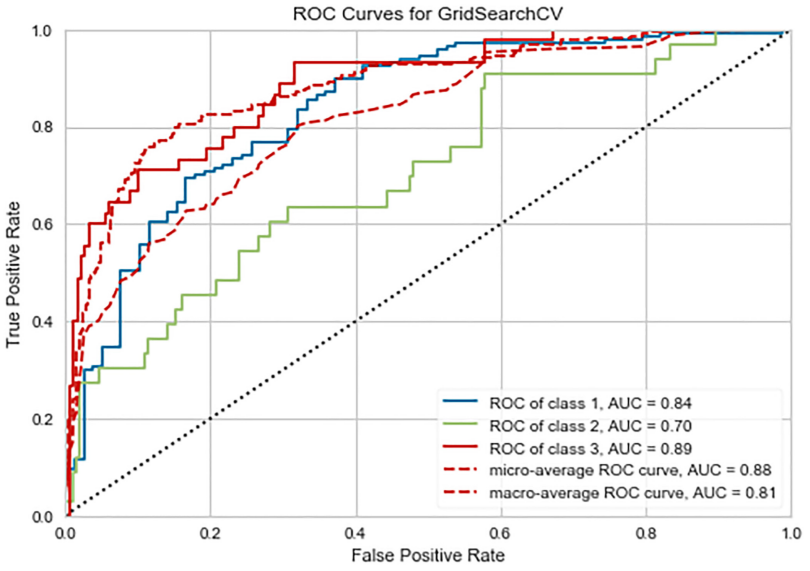


Figure 7.
ROC curve of logistic regression algorithm for Model 3

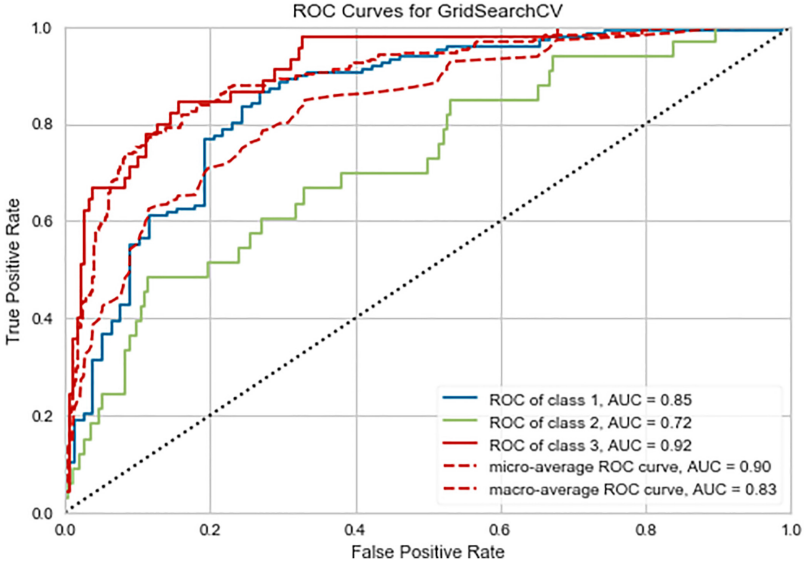


Figure 8.
ROC curve of logistic regression algorithm for Model 4

type of complaints are segregated and used as alternatives and weights of the criteria determined for each category of complaints are utilized to evaluate the ranking score of each tweet.

The results obtained after executing the steps in TOPSIS methods are explained below using some subset of tweets belonging to road, safety and noise pollution categories. Table 11 shows the example of the tweets for which the ranking need to be calculated using TOPSIS (see Tables 12–18).

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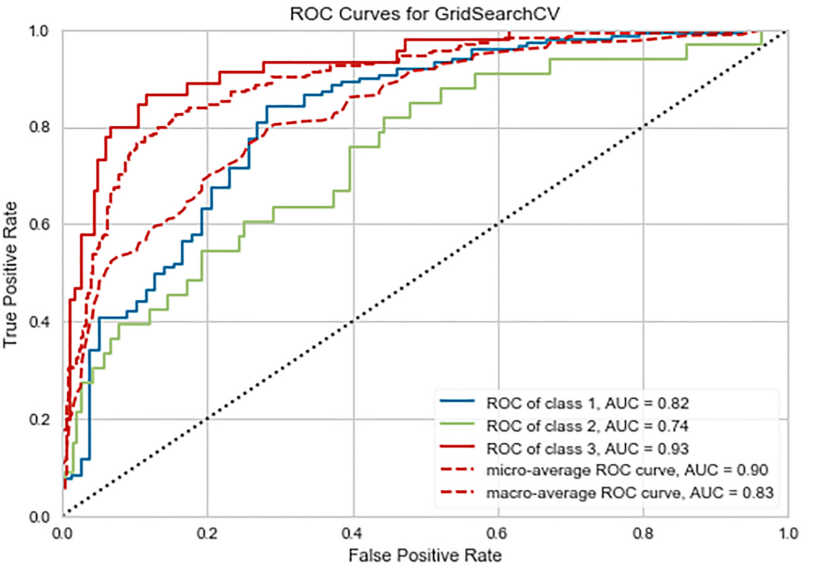


Figure 9.
ROC curve of logistic regression algorithm for Model 5

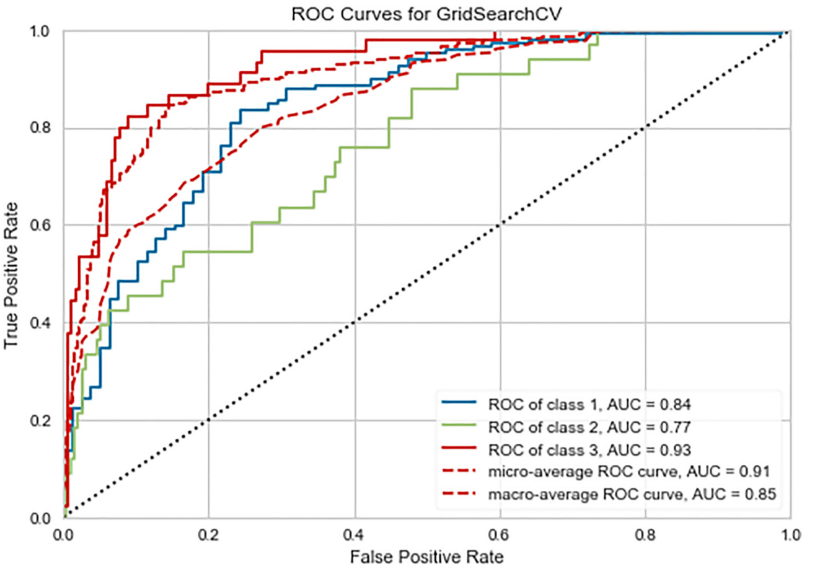


Figure 10.
ROC curve of logistic regression algorithm for Model 6

Importance	Description
1	Not important at all
2	Slight importance
3	Important
4	Moderate to strong importance
5	Strong importance

Table 7.
Scale used for pairwise comparison in the proposed study

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From the above result, the relative closeness to the ideal solution is maximum for tweet ID 2654 followed by tweet ID 275 and 7065, so the tweets are ranked in the same order as 1, 2 and 3.

Similarly, after following all the above mentioned steps, the matrices are derived for separation from ideal solution, NIS and relative closeness to ideal solution for each of the "Safety" and "Noise pollution" related tweets as follows. Based upon the relative closeness score, the tweets are ranked as 1, 2 or 3.

The tweets from table are rearranged based upon the calculated ranking as below.

The tweets ranked according to the relative closeness score in each of the categories clearly indicate that the combination of AHP and TOPSIS algorithms has shown promising results for prioritizing the urban issues category-wise. However, instead of individual category-wise ranking, tweets need to be ranked across the categories which will help to locate the critical issues immediately. So, to evaluate the overall ranking of the tweets, in addition to the category-wise relative closeness score, two more parameters are used, i.e. weight of "type of complaint" criteria and severity score predicted using the MCC approach as explained in the previous section.

5.2.2 Weights calculation of each type of complaint using AHP. While deciding the priority of the complaint, along with the content present in the text, the type of complaint also plays vital role. So by using the AHP method, pairwise CM for each type of complaint was formulated. As shown in Table 19 and accordingly, weights are derived for further

Road	Fear	ang_fru	req	day_type	Holiday	loc_present	day_night	Season	Gender
Fear	1	4/3	4/3	4/3	4/2	3/5	4/3	4/5	3/2
ang_fru	3/4	1	3/4	4/3	4/2	4/5	3/4	3/4	3/2
req	3/4	4/3	1	4/3	4/2	4/5	3/4	3/4	3/2
day_type	3/4	3/4	3/4	1	3/2	3/5	3/4	3/4	3/2
Holiday	2/4	2/4	2/4	2/3	1	2/5	2/3	2/3	3/2
loc_present	5/3	5/4	5/4	5/3	5/2	1	5/3	5/3	5/2
day_night	3/4	4/3	4/3	4/3	3/2	3/5	1	3/4	3/2
Season	5/4	4/3	4/3	4/3	3/2	3/5	4/3	1	3/2
Gender	2/3	2/3	2/3	2/3	2/3	2/5	2/3	2/3	1

Table 8. Pair-wise comparison of matrix for all criteria w.r.t. complaint type as "Road"

Criteria→ Type of complaint↓	Fear	ang_fru	req	day_type	Holiday	loc_present	day_night	Season	Gender
Road	0.13	0.11	0.11	0.09	0.07	0.17	0.11	0.13	0.07
Safety	0.15	0.08	0.15	0.09	0.08	0.13	0.11	0.09	0.13
Noise poll	0.07	0.17	0.17	0.11	0.10	0.15	0.14	0.07	0.03
Traffic	0.11	0.17	0.13	0.11	0.09	0.13	0.10	0.10	0.07
Train	0.10	0.10	0.13	0.10	0.11	0.13	0.11	0.11	0.14
Garbage	0.07	0.13	0.14	0.11	0.08	0.17	0.09	0.15	0.05
ill_parking	0.07	0.13	0.14	0.11	0.08	0.17	0.09	0.15	0.05
crime	0.15	0.07	0.14	0.09	0.08	0.13	0.12	0.09	0.13
Footpath	0.12	0.11	0.12	0.09	0.07	0.16	0.12	0.14	0.07
Encro	0.11	0.15	0.15	0.09	0.07	0.19	0.08	0.13	0.04
trans	0.11	0.15	0.15	0.09	0.07	0.19	0.08	0.13	0.04
ill_ban	0.13	0.14	0.15	0.11	0.09	0.17	0.10	0.08	0.04
inconv	0.10	0.13	0.13	0.08	0.06	0.17	0.07	0.12	0.13
Manhole	0.13	0.10	0.11	0.09	0.08	0.17	0.14	0.15	0.03

Table 9. Summarized criteria weights for individual type of complaints

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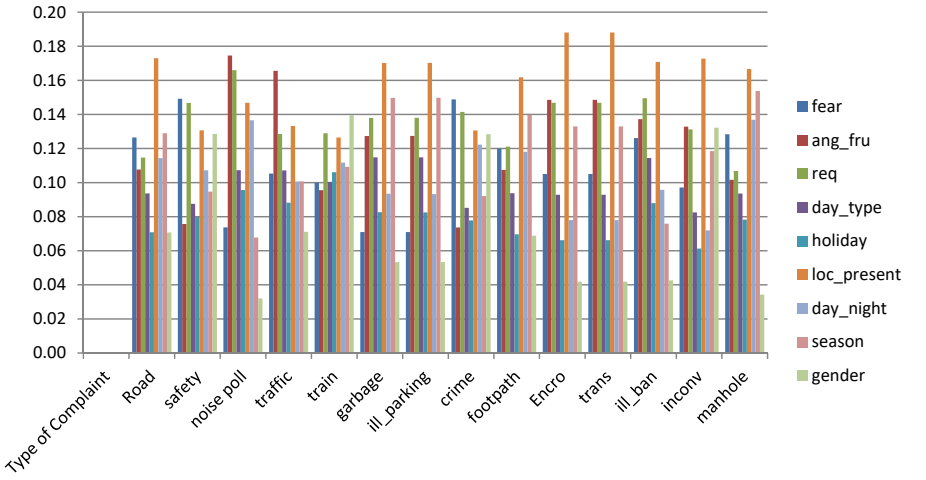


Figure 11.
Importance of each criterion for complaints after AHP evaluation

Type of complaint	λmax	CI	RI	CR
Road	9.120480	0.015060	1.45	0.010386
Safety	9.068013	0.008502	1.45	0.007163
Noise poll	9.097844	9.097844	1.45	0.012230
Traffic	3.963281	0.012281	1.45	0.008470
Train	9.007476	0.000934	1.45	0.000644
Garbage	9.341522	0.042690	1.45	0.029442
ill_parking	9.335981	0.041998	1.45	0.028964
Crime	9.076679	0.009585	1.45	0.006610
Footpath	9.101447	0.012681	1.45	0.008745
Encro	9.220310	0.027539	1.45	0.018992
Trans	9.220310	0.027539	1.45	0.018992
ill_ban	9.155715	0.019464	1.45	0.013424
inconv	9.243348	0.030419	1.45	0.020978
Manhole	9.158571	0.019821	1.45	0.013670

Table 10.
Consistency ratio for each type of complaints

processing. The final weights determined by applying the equations of the AHP method are as displayed in Table 20.

While examining the derived weights, we can clearly observe that the life-threatening issues, like crime, safety, manhole and road, have higher weights as compared to the nonemergency issues.

5.3 Evaluating overall ranking of tweets

In this section, the results obtained from MCC and MCDM methods are combined together to determine the overall ranking of the tweet across the categories of complaints. Final ranking of the tweet was calculated using equation (11) as mentioned in detail in the methodology section. Table 21 displays the tweets arranged is descending order of the priority score derived using the following parameters: relative closeness score, "type of comp" weight and severity score.

From the result displayed in Table 21, it is observed that even if first three tweets are classified with severity score as 3 which indicate immediate action is required, the derived

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Type of complaint	Tweet id	Tweet
Road	275	The bridge near Belapur station Navi Mumbai Maharashtra is in a dangerous condition. Plzz Share This Central Railway look in to this plzz @mybmc @RailwaySeva @WesternRly @CPMumbaiPolice @RailMinIndia
	2654	ANOTHER URGENCY. A big ditch right outside Manmala Devi Mandir, Mahim is causing cars lining up till Ganga Vihar circle. This road is already in a bad traffic zone. Pls attend immediately @mybmcRoads @mybmcWardGN @mybmc @MumbaiPolice @RoadsOfMumbai @mumbaitraffic @mumbaimatterz https://t.co/AoCi8HQUcI
	7065	@mybmcSWM @mybmcWardPN @mybmc @mybmcRoads @prabhu_suneel @mybmcSWD This is the road of Pal Nagar, where the condition has deteriorated a lot due to rain, travelers are suffering a lot, here many school children come and request to BMC and repair it as soo
Safety	7614	@mybmc @mybmcWardKE The pictured pole is not insulated properly resulting electrical shock on touching. Location MV Road next to Taxi stand opp Siddhivinayak Ganesh Temple, Andheri east. Plz resolve at the earliest. https://t.co/XxxhxxN4KG
	53	@MumbaiPolice people stand outside shweta beer shop (near fatima high school vidyavihar west) and they have their drinks there and some of them teasing women plz look into it. Its an request
	3995	@CPMumbaiPolice @AUThackeray @mybmcHealthDept @mybmcSWD @mybmcWardPS @RoadsOfMumbai @NBTMumbai Nitin Niwas, jawahar Ngr Rd.15, Goregaon (E), Kulal Dosa Center, Unsafe and illegal food is being sold, roads are being occupied, even in the rainy season
Noise pollution	3549	Its almost 12 [. . .] mid night [. . .] loud noise of dhol near Kajupada, Borivali east [. . .] please help @MumbaiPolice @CPMumbaiPolice
	6063	@mumbaipolice Is there a dead line during celebrations and festivals for playing music? Is 10:30 p.m. [. . .] and loud music is being played. @ st. Francis road, Bandra west
	7022	Its done by both the restaurant day and night by washing their utensils so loudly and their employees are really idiots, sir y are u not taking action, is their boss some kind of big shot?, we are staying at Alexandra terrace and already comp against this

Table 11.
Sample tweets of different categories like "road," "safety," "noise pollution"

Tweet id	Fear	ang_fru	req	day_type	Holiday	loc_present	day_night	Season	Gender
(weights)	*0.13*	*0.11*	*0.11*	*0.09*	*0.07*	*0.17*	*0.11*	*0.13*	*0.07*
275	1	0	1	2	2	1	1	2	1
2654	1	0	1	2	2	1	1	2	2
7065	0	0	1	1	2	1	2	2	1

Table 12.
Decision table for tweets belonging to "road" type of complaint along with the weights of the criteria derived using AHP method

ranking score helps to prioritize the most urgent tweet among the three immediate tweets. Similarly, the tweets with severity score as 2 indicating urgent tweets are also arranged according to the priority. The table only shows the subset of the large dataset used for the experimentation; however, the same framework is applied to determine the priority score of all tweets.

6. Conclusion

After applying the MCC on the tweets mentioning urban issues to identify immediate, urgent and moderate tweets, it is observed that only classification was not sufficient in

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Table 13.
Standard decision matrix for tweets belonging to "road" type of complaint

Tweet id	Fear	ang_fru	req	day_type	Holiday	loc_present	day_night	Season	Gender
(weights)	*0.13*	*0.11*	*0.11*	*0.09*	*0.07*	*0.17*	*0.11*	*0.13*	*0.07*
275	0.141421	0	0.096674	0.071247	0.07231	0.067884	0.0553	0.065407	0.0341
2654	0.141421	0	0.096674	0.071247	0.07231	0.067884	0.0553	0.065407	0.068199
7065	0	0	0.096674	0.035624	0.07231	0.067884	0.1106	0.065407	0.0341

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Tweet id	Fear	ang_fru	req	day_type	Holiday	loc_present	day_night	Season	Gender
(weights)	*0.13*	*0.11*	*0.11*	*0.09*	*0.07*	*0.17*	*0.11*	*0.13*	*0.07*
275	0.017876	0	0.011085	0.006671	0.005123	0.011738	0.006322	0.008431	0.00241
2654	0.017876	0	0.011085	0.006671	0.005123	0.011738	0.006322	0.008431	0.004819
7065	0	0	0.011085	0.003336	0.005123	0.011738	0.012643	0.008431	0.00241

Note(s): Positive ideal solution = {0.0178764, 0.0082083, 0.0110851, 0.0066713, 0.0051227, 0.0117382, 0.0126433, 0.0084311, 0.0048194}
Negative ideal solution = {0, 0, 0, 0, 0,0, 0, 0.004215,0.00241}

Table 14.
Weighted decision matrix for tweets belonging to "Road" type of complaint

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order to solve the issues using optimum resources. The tweets need to be ranked based upon their severity of issue in order to handle it within time before any disaster happens. However, the priority of the tweet depends upon various criteria whose different permutation combination can affect the ranking of the tweets. After a detailed literature survey, we observed MCDM methods were most appropriate in these scenarios and hence explored two most popular techniques, i.e. AHP for criteria weight evaluation and TOPSIS for the final ranking score of tweets. This proposed approach would enable local government to discover the limitation in the existing system to handle urban issues and demonstrate how several indicators explored during the analysis play a crucial role in handling them more promptly resulting in a better quality of life for citizens. The weight estimation and ranking methods employed in this study generated the result which clearly signifies that the proposed approach is easy to understand and reliable to rank complaint tweets based on their severity level helping the policymakers to manage with the limited resources. Results obtained have clearly ranked the life-threatening and safety-related tweets first as compared to the less urgent inconvenience-related complaints. Utilizing multiple parameters while deciding the urgency of the complaint tweets and applying the ensemble approach of MCC and MCDM for their ranking is the novelty of this study making local government smarter and more efficient.

One of the main limitations of the both methods is high computational complexity. As the number of tweets or number of types of complaints and derived features chosen as criteria increases, the computations also increase which may be slight time-consuming. However generally, from huge amount of extracted tweets, very few may be classified as urgent and immediate as compared to large amount of moderate tweets. Additionally, immediate or urgent tweets only discuss the life-threatening or disaster situations for

Table 15.
Separation from ideal solution, negative ideal solution and relative closeness to ideal solution for "road" tweets

Tweet id	S^*	S'	$S'/(S^* + S')$	Rank
275	0.010637	0.026621	0.71450	2
2654	0.010360	0.02673	0.72067	1
7065	0.020096	0.021809	0.52043	3

Note(s): Where S^* = Separation from ideal solution, S' = Separation from negative ideal solution and $S'/(S^* + S')$ = Relative closeness to the ideal solution

Table 16.
Separation from ideal solution, negative ideal solution and relative closeness to ideal solution for "Safety tweets"

Tweet id	S^*	S'	$S'/(S^* + S')$	Rank
7614	0.01534	0.03165	0.673542	2
53	0.009026	0.034846	0.794266	1
3995	0.028613	0.021841	0.432891	3

Table 17.
Separation from ideal solution, negative ideal solution and relative closeness to ideal solution for "Noise pollution tweets"

Tweet id	S^*	S'	$S'/(S^* + S')$	Rank
3549	0.073735	0.038225	0.341419	1
6063	0.079695	0.018758	0.19053	2
7022	0.080559	0.017371	0.177379	3

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Type of complaint	Tweet id	Tweet	Relative closeness score	Ranking
Road	2654	ANOTHER URGENCY. A big ditch right outside Manmala Devi Mandir, Mahim is causing cars lining up till Ganga Vihar circle. This road is already in a bad traffic zone. Pls attend immediately @mybmcRoads @mybmcWardGN @mybmc @MumbaiPolice @RoadsOfMumbai @mumbaitraffic @mumbaimatterz https://t.co/AoCi8HQUcI	0.72067	1
	275	The bridge near Belapur station Navi Mumbai Maharashtra is in a dangerous condition. Plzz Share This Central Railway look in to this plzz @mybmc @RailwaySeva @WesternRly @CPMumbaiPolice @RailMinIndia	0.71450	2
	7065	@mybmcSWM @mybmcWardPN @mybmc @mybmcRoads @prabhu_suneel @mybmcSWD This is the road of Pal Nagar, where the condition has deteriorated a lot due to rain, travelers are suffering a lot, here many school children come and request to BMC and repair it as soo	0.52043	3
Safety	53	@MumbaiPolice people stand outside shweta beer shop (near fatima high school vidyavihar west) and they have their drinks there and some of them teasing women plz look into it. Its an request	0.794266	1
	7614	@mybmc @mybmcWardKE The pictured pole is not insulated properly resulting electrical shock on touching. Location MV Road next to Taxi stand opp Siddhivinayak Ganesh Temple, Andheri east. Plz resolve at the earliest. https://t.co/XxxhxxN4KG	0.673542	2
	3995	@CPMumbaiPolice @AUThackeray @mybmcHealthDept @mybmcSWD @mybmcWardPS @RoadsOfMumbai @NBTMumbai Nitin Niwas, jawahar Ngr Rd.15, Goregaon (E), Kulal Dosa Center, Unsafe and illegal food is being sold, roads are being occupied, even in the rainy season	0.432891	3
Noise pollution	3549	Its almost 12 [. . .] mid night [. . .] loud noise of dhol near Kajupada, Borivali east [. . .] please help @MumbaiPolice @CPMumbaiPolice	0.341419	1
	6063	@mumbaipolice Is there a dead line during celebrations and festivals for playing music? Is 10:30p.m. [. . .] and loud music is being played. @ st. Francis road, Bandra west	0.19053	2
	7022	Its done by both the restaurant day and night by washing their utensils so loudly and their employees are really idiots, sir y are u not taking action, is their boss some kind of big shot?, we are staying at Alexandra terrace and already comp against this. https://t.co/p3QsV5V1o0	0.177379	3

Table 18.
Category-wise ranking of tweets

which the priority should be decided. Hence, in case of high computational complexity, only immediate or urgent tweets will be ranked, and based on that, the local government can take time-bound actions and save life.

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Table 19.
Pair-wise comparison matrix for "type of complaint" criteria

	Traffic	Road	Train	Garbage	ille_park	Crime	Safety	Footpath	Noise poll	encr	Transp	ill_ban	inconv	Manhole
Traffic	1	3/5	4/3	4/3	4/3	3/5	3/5	4/3	4/3	4/3	4/3	4/3	4/3	4/5
Road	5/3	1	5/3	5/3	5/3	4/5	4/5	4/3	4/3	4/3	4/3	4/3	4/3	4/5
Train	3/4	3/5	1	4/3	4/3	3/5	3/5	4/3	4/3	4/3	4/3	4/3	4/3	3/5
Garbage	3/4	3/5	3/4	1	3/4	3/5	3/5	3/4	3/4	3/4	3/4	4/3	3/4	3/5
ille_park	3/4	3/5	3/4	4/3	1	3/5	3/5	3/4	4/3	4/3	4/3	4/3	3/4	3/5
Crime	5/3	5/4	5/3	5/3	5/3	1	4/5	5/3	5/3	5/3	5/3	5/3	5/3	5/4
Safety	5/3	5/4	5/3	5/3	5/3	5/4	1	5/3	5/3	5/3	5/3	5/3	5/3	5/4
Footpath	3/4	3/4	3/4	4/3	4/3	3/5	3/5	1	4/3	4/3	4/3	3/4	4/3	3/5
Noise poll	3/4	3/4	3/4	4/3	3/4	3/5	3/5	3/4	1	4/3	3/4	3/4	3/4	3/5
encr	3/4	3/4	3/4	4/3	3/4	3/5	3/5	3/4	3/4	1	3/4	3/4	3/4	3/5
transp	3/4	3/4	3/4	4/3	3/4	3/5	3/5	3/4	4/3	4/3	1	4/3	3/4	3/5
ille_ban	3/4	3/4	3/4	3/4	3/4	3/5	3/5	4/3	4/3	4/3	3/4	1	3/4	3/5
inconv	3/4	3/4	3/4	4/3	4/3	3/5	3/5	3/4	4/3	4/3	4/3	4/3	1	3/5
Manhole	5/4	5/4	5/3	5/3	5/3	4/5	4/5	5/3	5/3	5/3	5/3	5/3	5/3	1

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6.1 Recommendations

The proposed system can be used to prioritize the urban issues raised in different cities across India. Continuous update in the list of keywords used to extract the most relevant tweets can help in reducing the noisy tweets and improving the overall evaluation performance. Further neural network algorithms can be explored for improved classification performance. Along with that, considering the imbalanced distribution of the immediate, urgent and moderate tweets, various

Table 20. Weights pertaining to each "type of complaint"

Traffic	Road	Train	Garbage	ille_park	Crime	Safety	Footpath	Noise poll	encr	Transp	ille_ban	inconv	Manhole
0.07	0.09	0.07	0.05	0.06	0.10	0.10	0.07	0.06	0.05	0.06	0.06	0.06	0.10

Table 21. Final ranking calculation of tweets

Type	Tweet id	Tweet	Relative closeness score	"Type of comp" weight	Severity score from multi-class classification	Final rank
Safety	7614	@mybmc @mybmcWardKE The pictured pole is not insulated properly resulting electrical shock on touching. Location MV Road next to Taxi stand opp Siddhivinayak Ganesh Temple, Andheri east. Plz resolve at the earliest	0.673541	0.1	3	0.202062
Road	2654	ANOTHER URGENCY. A big ditch right outside Manmala Devi Mandir, Mahim is causing cars lining up till Ganga Vihar circle. This road is already in a bad traffic zone. Pls attend immediately @mybmcRoads @mybmcWardGN @mybmc @MumbaiPolice @RoadsOfMumbai @mumbaitraffic @mumbaimatterz	0.720673	0.09	3	0.187252
Safety	53	@MumbaiPolice people stand outside shweta beer shop (near fatima high school vidyavihar west) and they have their drinks there and some of them teasing women plz look into it. Its an request	0.794266	0.1	2	0.158853
Road	275	The bridge near Belapur station Navi Mumbai Maharashtra is in a dangerous condition. Plzz Share This Central Railway look in to this plzz @mybmc @RailwaySeva @WesternRly @CPMumbaiPolice @RailMinIndia	0.714508	0.09	2	0.123766

(*continued*)

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Type	Tweet id	Tweet	Relative closeness score	"Type of comp" weight	Severity score from multi-class classification	Final rank
Noise pollution	3549	Its almost 12[. . .]mid night[. . .] loud noise of dhol near Kajupada, Borivali east [. . .] please help @MumbaiPolice @CPMumbaiPolice	0.341418	0.06	3	0.056491
Road	7065	@mybmcSWM @mybmcWardPN @mybmc @mybmcRoads @prabhu_suneel @mybmcSWD. This is the road of Pal Nagar, where the condition has deteriorated a lot due to rain, travelers are suffering a lot, here many school children come and request to BMC and repair it as soon as posible	0.520437	0.09	1	0.045074
Safety	3995	@CPMumbaiPolice @AUThackeray @mybmcHealthDept @mybmcSWD @mybmcWardPS @RoadsOfMumbai @NBTMumbai Nitin Niwas, jawahar Ngr Rd.15, Goregaon (E), Kulal Dosa Center, Unsafe and illegal food is being sold, roads are being occupied, even in the rainy season	0.432890	0.1	1	0.043289
Noise pollution	6063	@mumbaipolice Is there a dead line during celebrations and festivals for playing music? Is 10:30p.m. [. . .] and loud music is being played. @ st. Francis road, Bandra west	0.19053	0.06	2	0.021016
Noise pollution	7022	Its done by both the restaurant day and night by washing their utensils so loudly and their employees are really idiots, sir y are u not taking action, is their boss some kind of big shot?, we are staying at Alexandra terrace and already comp against this	0.177379	0.06	1	0.009783

Table 21.

techniques for handling the imbalanced dataset like under-sampling, over-sampling and synthetic minority oversampling technique (SMOTE) will help in building the unbiased classifiers and probably produce more accurate results. Also, other MCDM methods like ANP, ENTROPY, ELECTRE and VIKOR etc. may be explored, and the results obtained can be compared with the results achieved in this study.

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